

BPM 1
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Cholinergic Agonists and Antagonists

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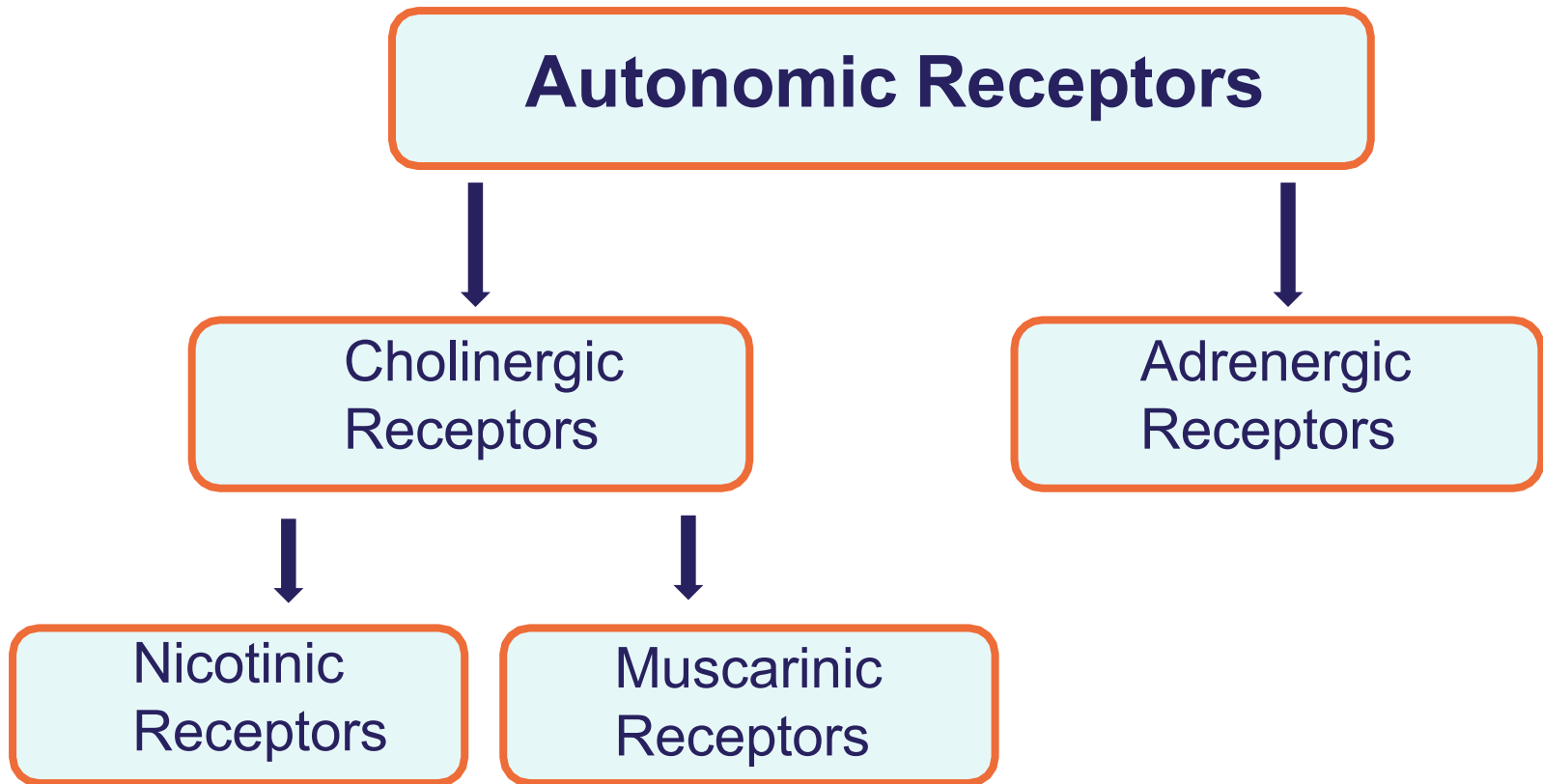


Objectives

SOM.MK.1.BPM1.1.FTM.4.PHAR.0040	List the different classes of cholinergic drugs.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0041	Describe the muscarinic and nicotinic actions of acetylcholine.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0042	Describe the two classes of direct-acting cholinergic agonists. Describe the organ system effects, mechanism of action, uses and adverse effects of muscarinic receptor agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0043	Describe the organ system effects, mechanism of action, uses and adverse effects of nicotinic receptor agonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0044	Describe the organ system effects, mechanism of action, uses and adverse effects of indirect-acting cholinergic agonists (anticholinesterases). Compare and contrast the effects of direct-acting and indirect-acting cholinergic agonists on the cardiovascular system.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0045	List the different classes of cholinergic antagonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0046	Describe the organ system effects, mechanism of action, therapeutic uses, adverse effects and contraindications of muscarinic receptor antagonists.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0047	Describe the mechanism of action of the ganglion blockers and predict their effects by applying knowledge of the predominant tone of a dually-innervated organ.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0048	Describe the mechanism of action, uses and main adverse effects of the nondepolarizing and the depolarizing neuromuscular blockers.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0049	Describe the mechanism of action and uses of botulinum toxin.



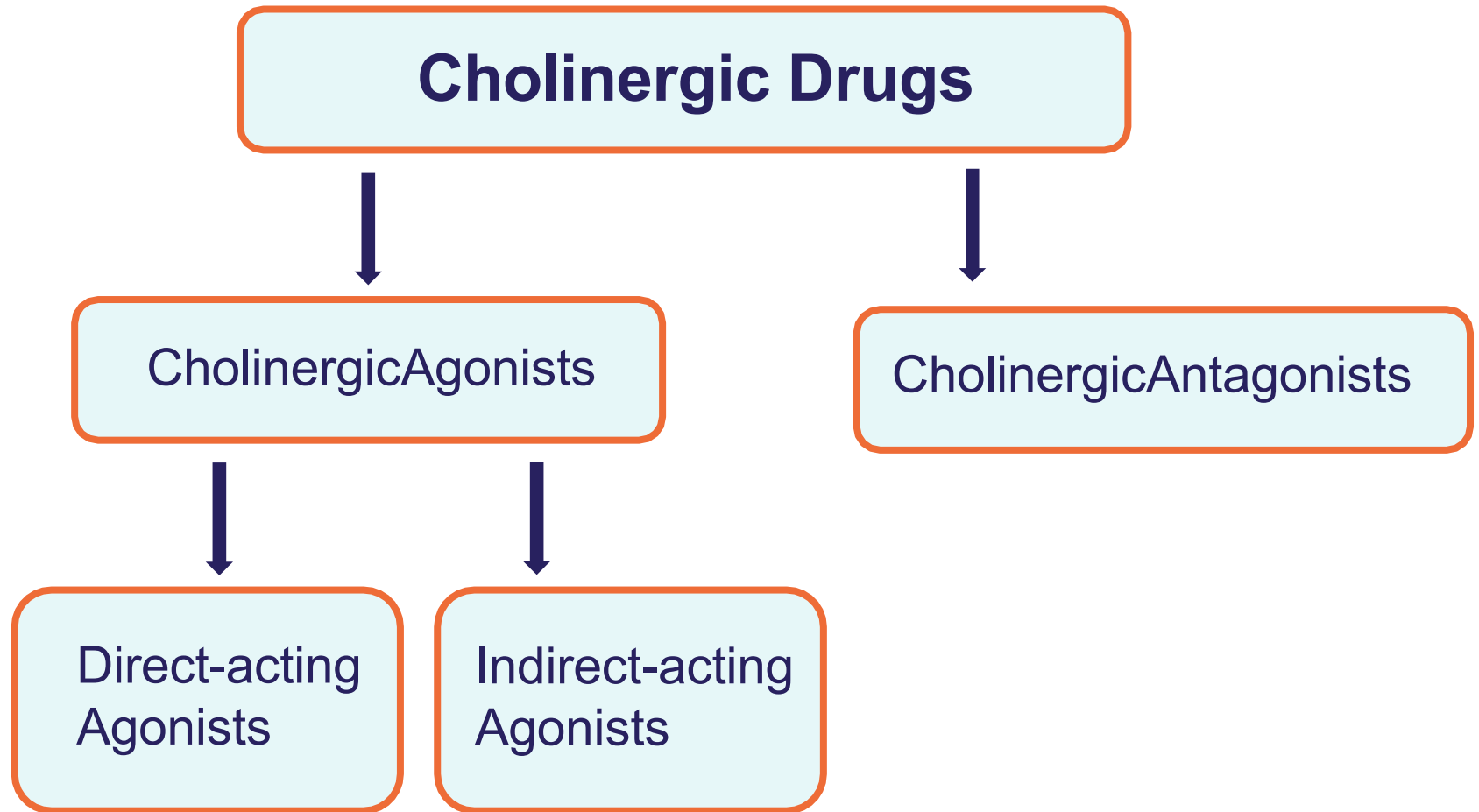
Autonomic and Cholinergic Receptors



CHOLINERGIC RECEPTORS

REC	SIGNALING	LOCATIONS	EFFECTS
M₁	↑ IP ₃ & DAG ↑ Ca ²⁺ Couples via G _q	Ganglia	Slow depolarization
		Vomiting center	Emesis
M₂	Opening of K ⁺ channels, ↓ cAMP Couples via G _i	Heart (mainly atria)	Decrease in heart rate
		Presynaptic nerve terminals	Inhibition of ACh release
M₃	↑ IP ₃ & DAG ↑ Ca ²⁺ Couples via G _q	Smooth muscle (Iris sphincter, ciliary, bronchial, gastrointestinal, genitourinary)	Contraction
		Secretory glands	Secretion
		Vascular endothelium (not innervated)	Vasodilation (via NO)
N_N	Opening of cation channels, depolarization	Autonomic ganglia	Ganglionic transmission
		Adrenal medulla	Secretion of Epi, NE
N_M	Opening of cation channels, depolarization	Neuromuscular junction	Neuromuscular transmission

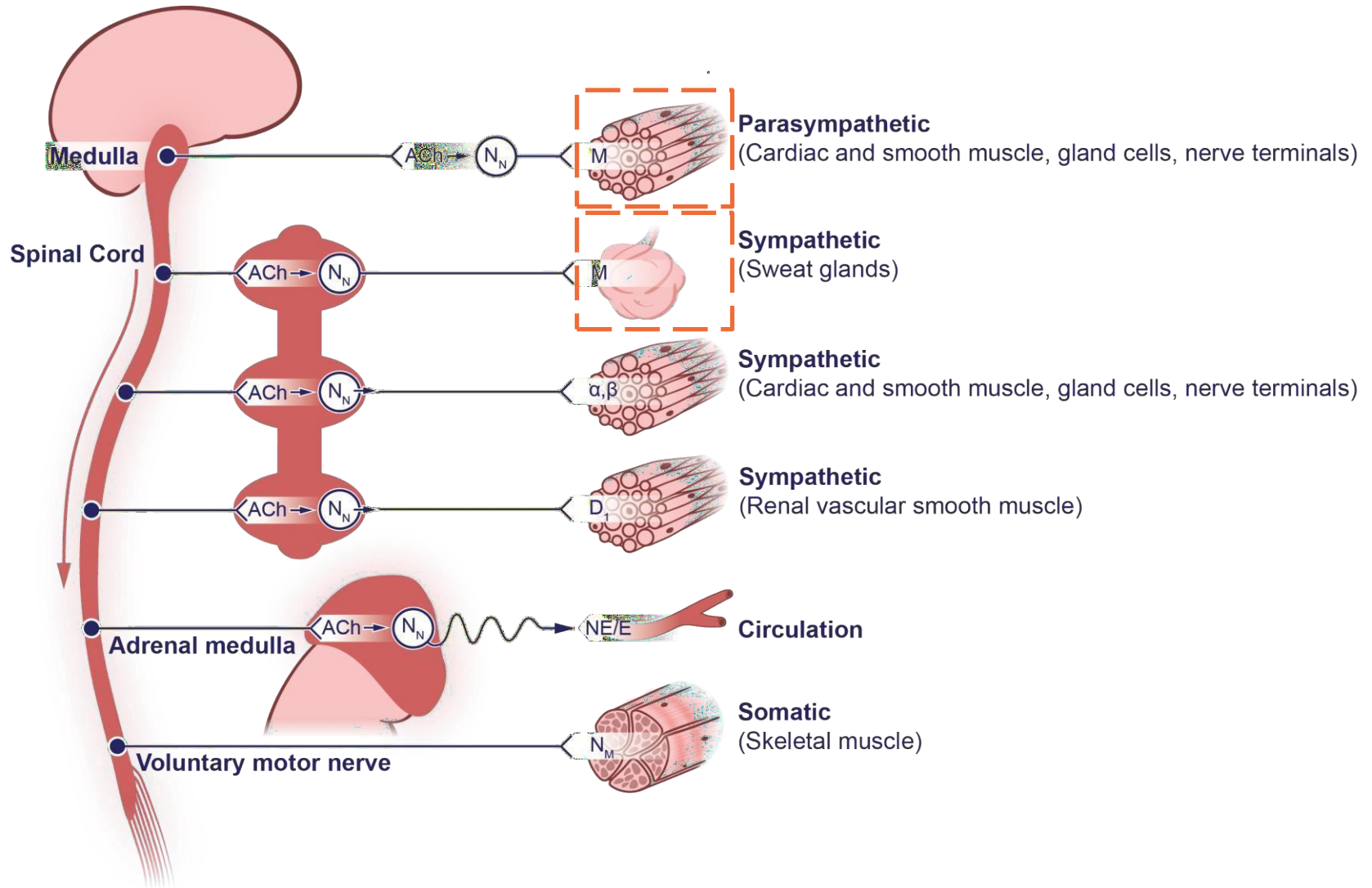
Cholinergic Drugs



Cholinergic Agonists

- **Direct-acting agents:** bind to and *activate* muscarinic or nicotinic receptors
 - Some are **selective** for the **muscarinic** *or* for the **nicotinic** receptor
 - Many have **effects on *both* receptors**
 - Most therapeutically useful drugs *preferentially* activate **muscarinic receptors**
- **Indirect-acting agents:** *inhibit* acetylcholinesterase

Direct-acting Cholinergic Agonists



Organ System Effects of Direct-acting Cholinergic Agonists

Cardiovascular Effects of ACh

Direct Effects of Acetylcholine

M3 effect: • Vasodilation (*via NO formation*)

M2 effect: • ↓ heart rate
• ↓ rate of conduction in the SA and AV nodes
• ↓ force of contraction

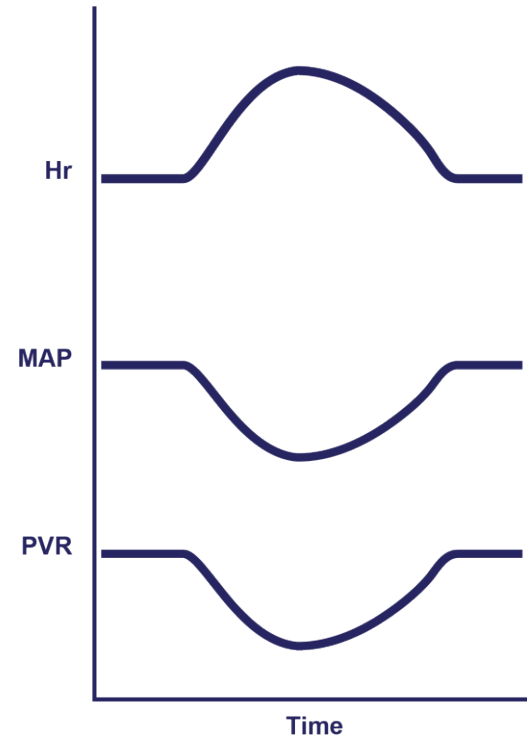
Some of these direct effects can be obscured by **baroreceptor reflexes**

Organ System Effects of Direct-acting Cholinergic Agonists

Cardiovascular Effects of ACh

IV injection of a **small dose** of acetylcholine produces a fall in blood pressure due to vasodilation (**M3 effect**) usually accompanied by **reflex tachycardia**.

ACh low dose

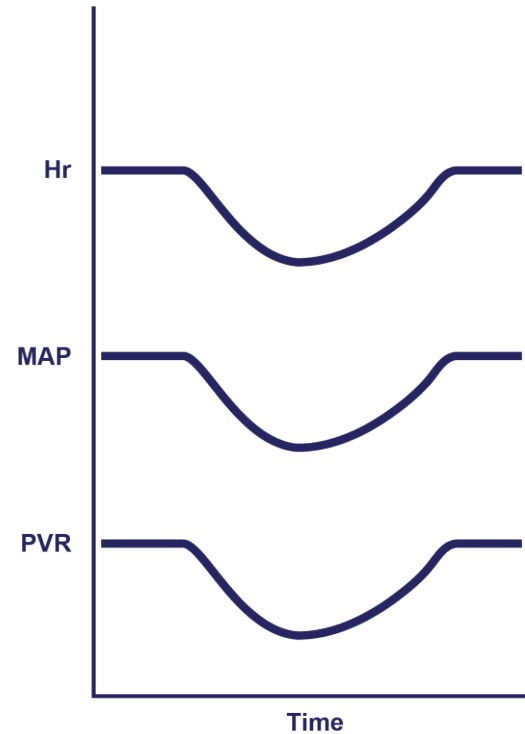


Organ System Effects of Direct-acting Cholinergic Agonists

Cardiovascular Effects of ACh

IV injection of a **large dose** of acetylcholine causes vasodilation, a fall in blood pressure (**M3 effect**) and **bradycardia** (**M2 effect**).

ACh high dose

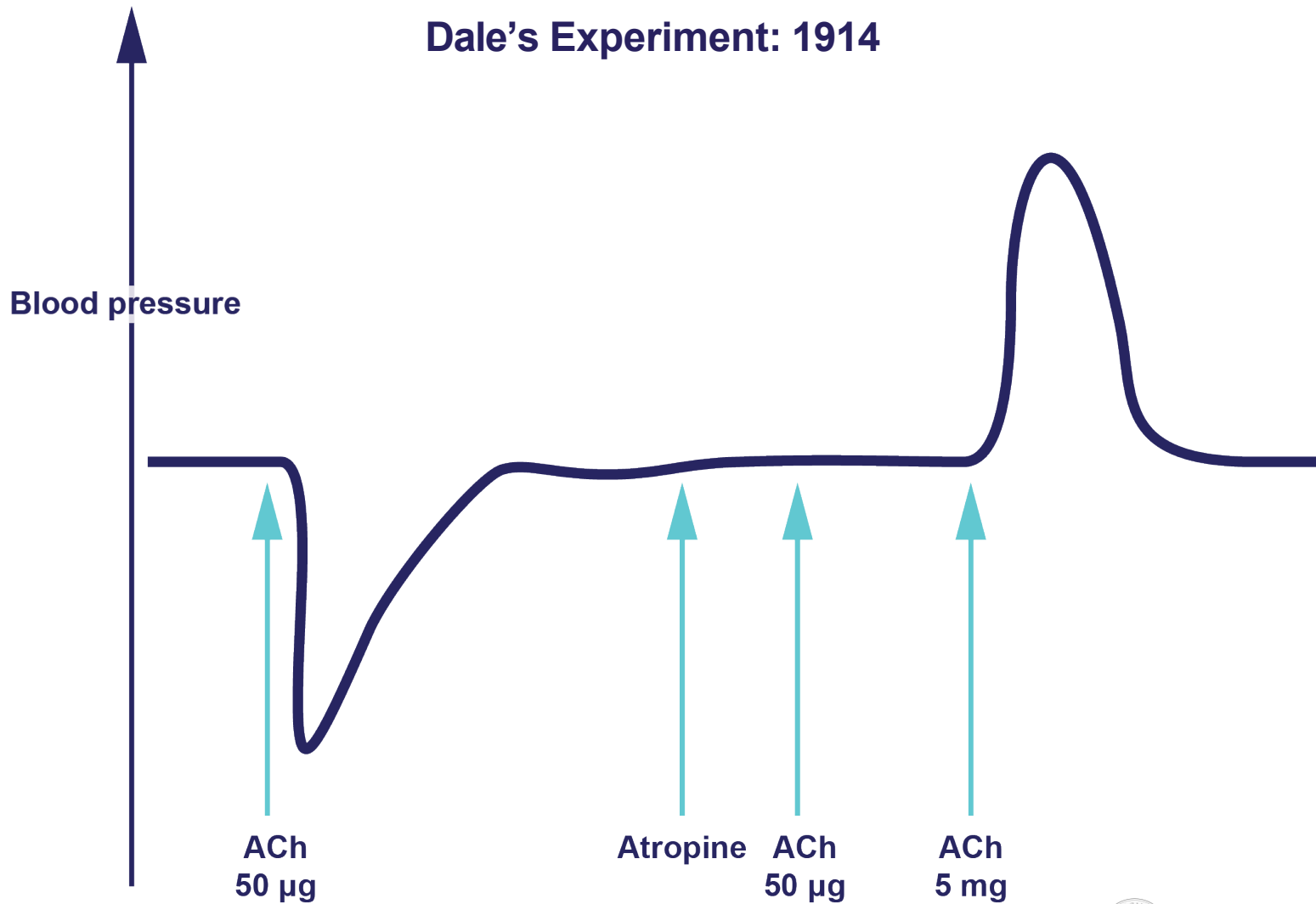


Organ System Effects of Direct-acting Cholinergic Agonists

Effects of Acetylcholine on Various Organ Systems

Rec	Tissue	Effect
M₃	Vasculature (endothelial cells)	Release of NO and vasodilation, decrease in BP
M₃	Eye iris	Miosis
M₃	Ciliary muscle	Accommodation of lens to near vision
M₃	Salivary/sweat/lacrimal glands	↑ Secretions
M₃	Bronchi	Constriction
M₂	Heart	↓ Heart rate, ↓ conduction velocity
M₃	GI tract	↑ Tone, peristaltic activity & secretions; relaxation of sphincters
M₃	Urinary bladder	Contraction of detrusor muscle; relaxation of sphincter

Nicotinic Effects of ACh

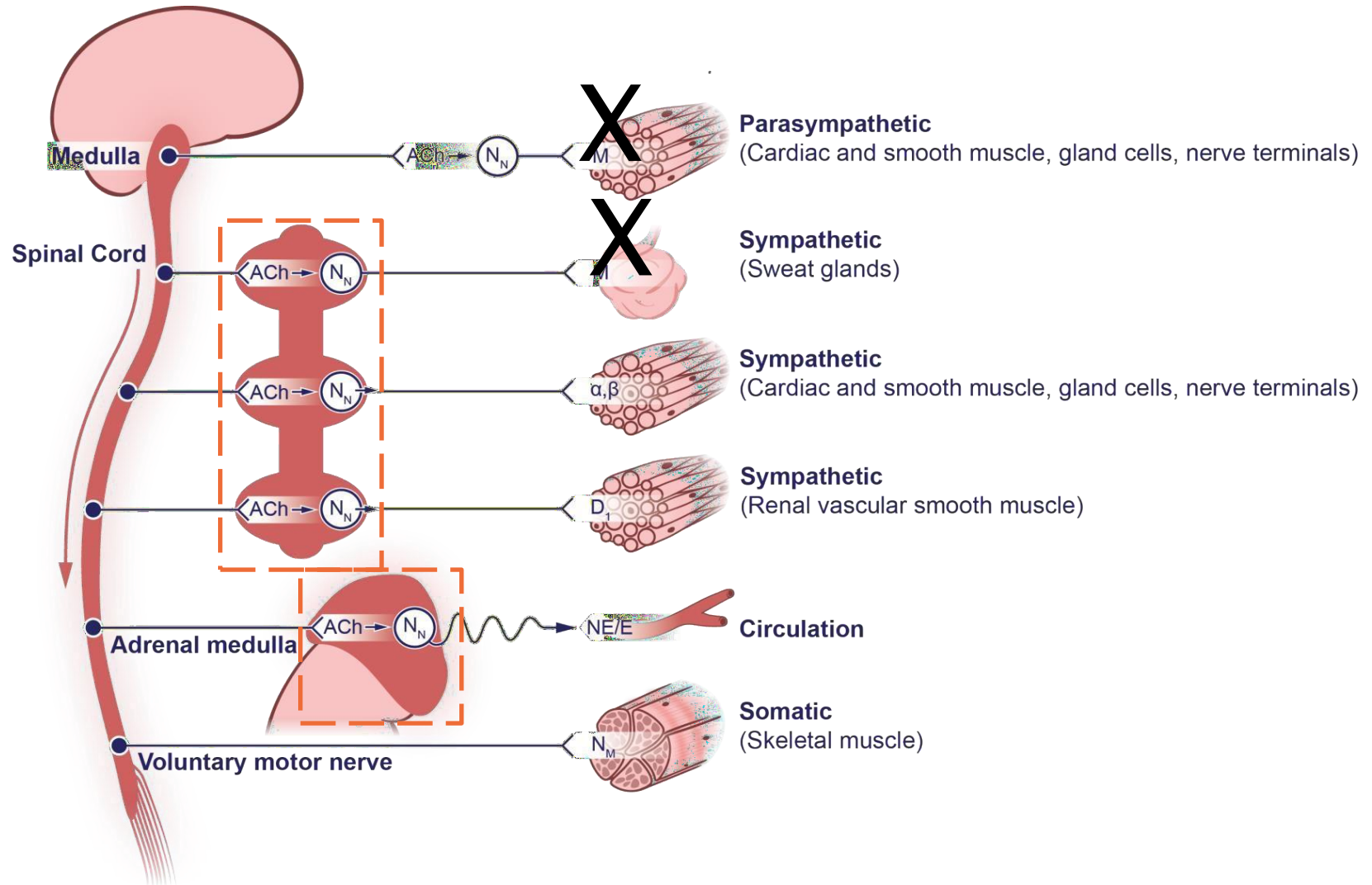


Nicotinic Effects of ACh

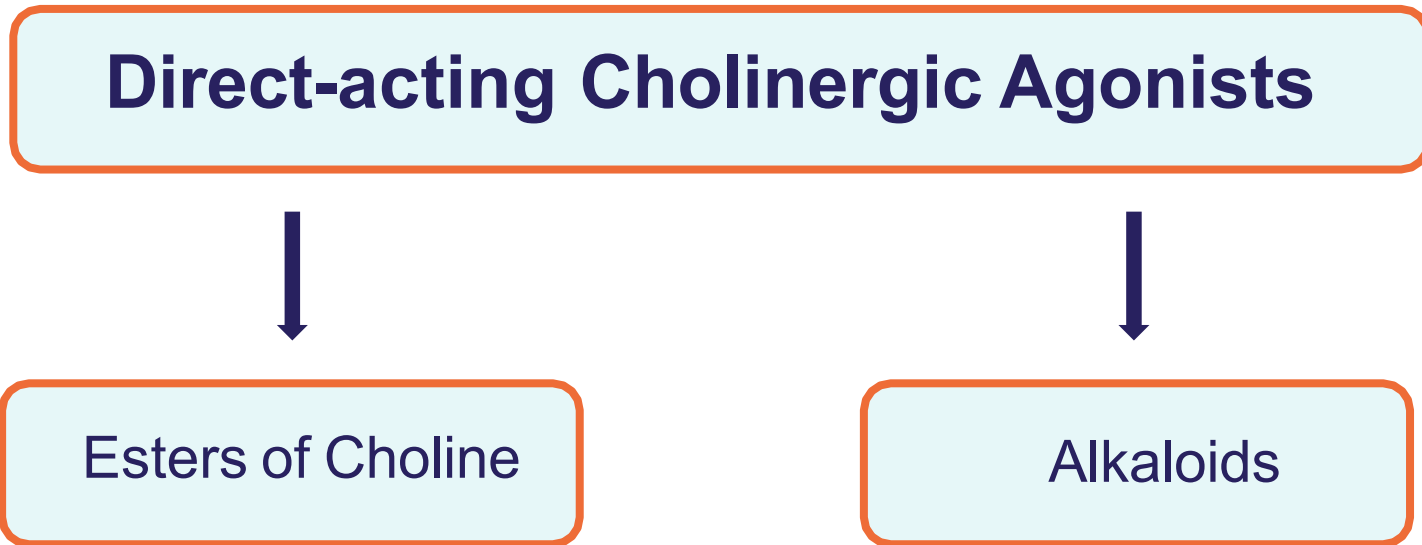
If muscarinic effects are *blocked* by a muscarinic antagonist such as **atropine**, large doses of acetylcholine produce **nicotinic effects**:

- Vasoconstriction
 - Increase in blood pressure
-
- These effects are due to ***stimulation*** of **sympathetic ganglia** and ***release*** of **epinephrine** from the **adrenal medulla**

Nicotinic Effects of ACh



Direct-acting Cholinergic Agonists



Choline Esters

Acetylcholine, Methacholine, Bethanechol

- Choline esters are **quaternary ammoniums**
- *Poorly* absorbed and *poorly* distributed into the CNS
- *Differ* in susceptibility to hydrolysis by cholinesterase
- Acetylcholine is very *rapidly* hydrolyzed
- **Methacholine** and **bethanechol** are more resistant to hydrolysis by cholinesterase

Choline Esters

Acetylcholine

- Muscarinic and nicotinic agonist

Use: To obtain miosis after delivery of the lens in cataract surgery, and other procedures where rapid miosis is required

Methacholine

- Muscarinic agonist

Use: Diagnosis of bronchial airway hyperreactivity in subjects who *do not* have clinically apparent asthma

Bethanechol

- Muscarinic agonist

Use: Post-operative urinary retention, and atony of the urinary bladder

Muscarinic Agonists: Adverse Events

The adverse effects mimic the effects of generalized cholinergic stimulation:

- Sweating
- Salivation
- Flushing
- Low blood pressure
- Nausea
- Abdominal pain
- Diarrhea
- Bronchospasm

Natural Alkaloids: Pilocarpine & Nicotine

Pilocarpine

- *Partial* muscarinic agonist
- Tertiary amine
- Stable to hydrolysis by acetylcholinesterase

Use:

- Glaucoma
- Treatment of dry mouth due to radiotherapy for cancer of head and neck

Adverse Effects:

- Like choline esters

Natural Alkaloids: Pilocarpine & Nicotine

Nicotine

- Agonist at **nicotinic receptors**
- Tertiary amine
- Depending on the dose, nicotine depolarizes autonomic ganglia, resulting first in stimulation and then paralysis

Actions: **Low doses:** ganglionic stimulation by depolarization
High doses: ganglionic blockade and neuromuscular blockade

Use: Smoking cessation therapy

Adverse Effects: **Low doses:** Response resembles simultaneous discharge of both parasympathetic and sympathetic nervous systems
High doses: acts like an antagonist

Indirect-acting Cholinergic Agents (Anticholinesterases)

Edrophonium: binds **reversibly** to the active site of the enzyme
- Inhibition is **short-lived**

Carbamates: form a covalent bond with the enzyme

- Physostigmine
- Neostigmine
- Pyridostigmine

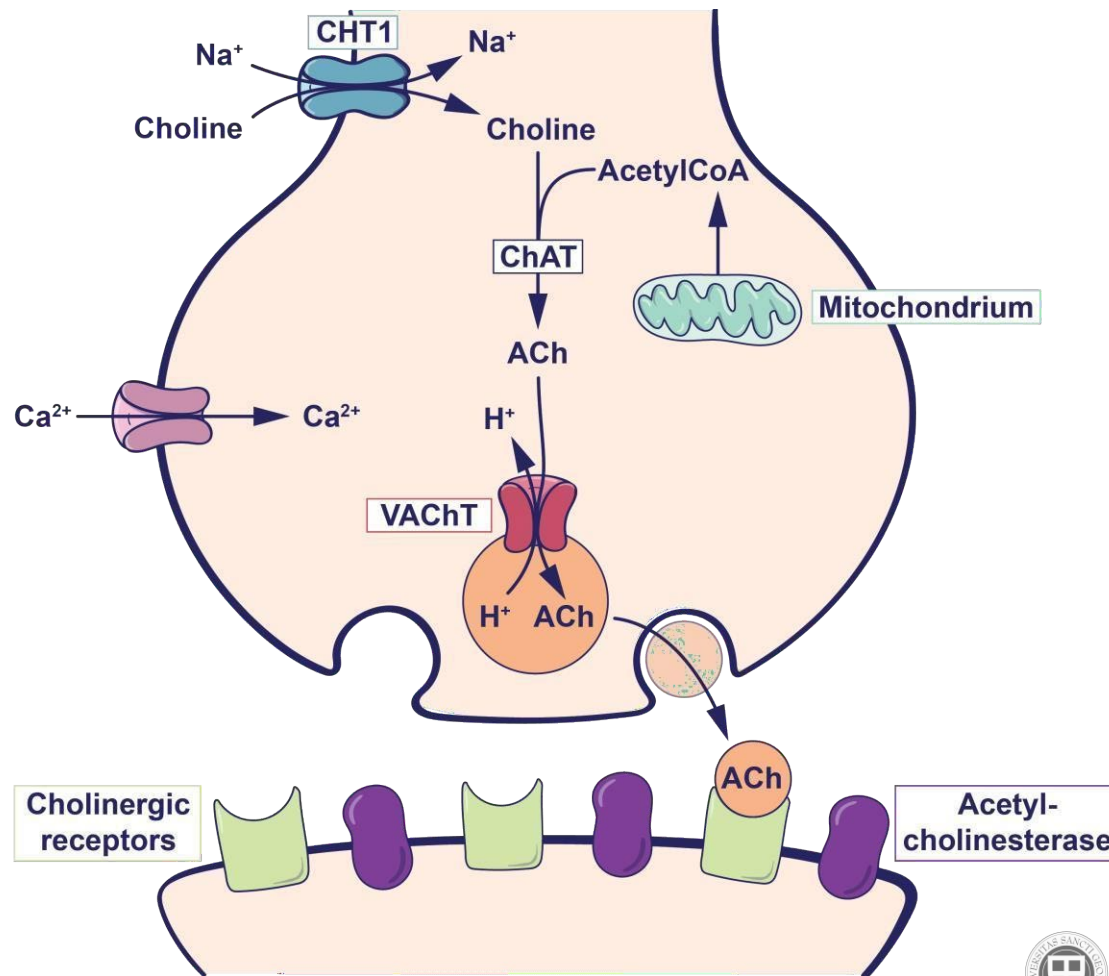
Organophosphates: phosphorylate the enzyme.

The covalent bond formed is extremely stable and hydrolyzes very slowly

- Malathion
- Sarin

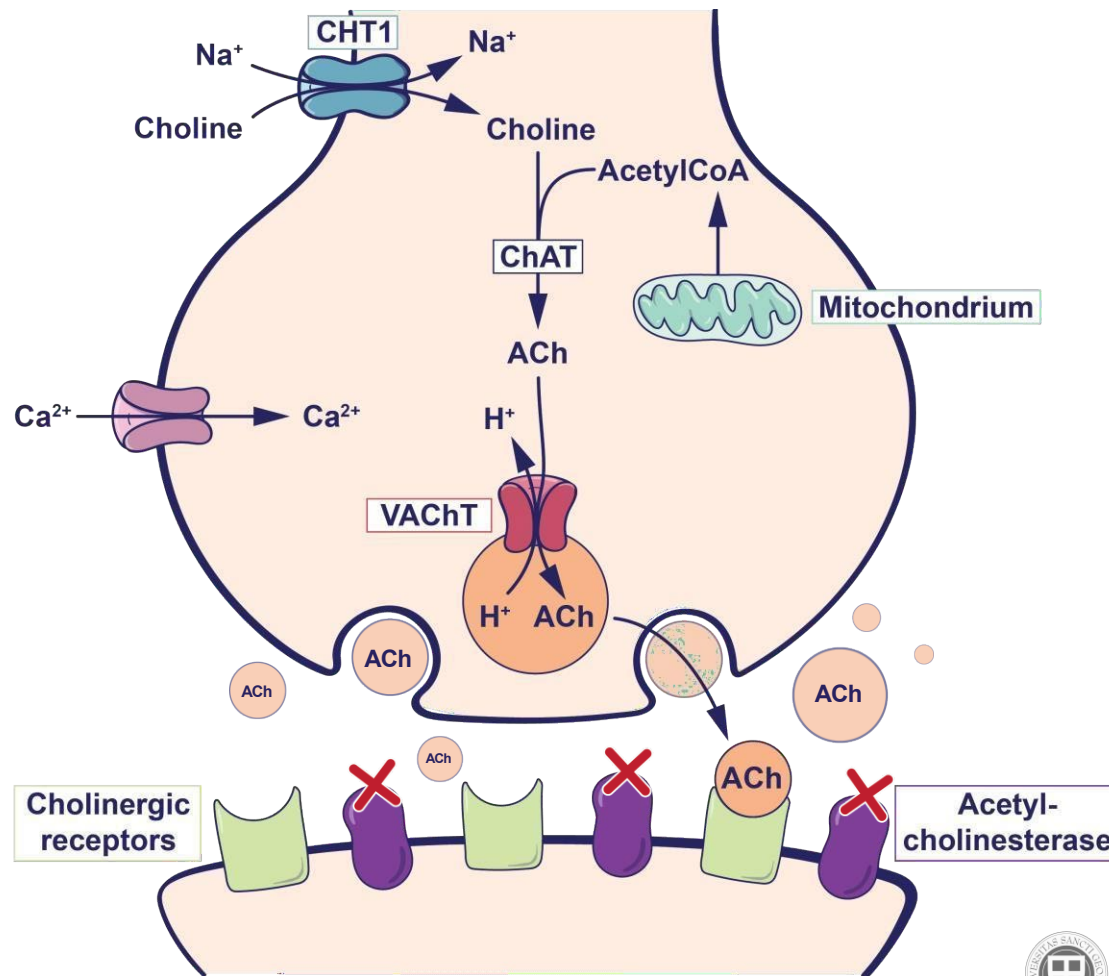
Indirect-acting Cholinergic Agents (Anticholinesterases)

MOA: Cholinesterase inhibitors **act by inhibiting acetylcholinesterase:** they increase concentration of endogenous acetylcholine



Indirect-acting Cholinergic Agents (Anticholinesterases)

MOA: Cholinesterase inhibitors **act by inhibiting acetylcholinesterase:** they increase concentration of endogenous acetylcholine



Anticholinesterases: Organ System Effects

Anticholinesterases amplify the action of endogenous acetylcholine

- Therefore, their effects are similar (***but not always identical***) to the effects of the direct-acting cholinomimetic agonists

Effects on the Cardiovascular System:

- **Vascular smooth muscle:** cholinesterase inhibitors have *minimal effects* because most vascular beds **lack cholinergic innervation**

Effects on the Neuromuscular Junction:

- Cholinesterase inhibitors *increase strength* of contraction
- Useful to reverse action of nondepolarizing neuromuscular blockers
- Useful in myasthenia gravis

Properties of Individual Anticholinesterases

Edrophonium, Carbamates & Organophosphates

Drug	Properties	Use
Edrophonium	• Quaternary ammonium • <i>Does not</i> enter CNS	• <i>Diagnosis of myasthenia gravis</i>: Edrophonium IV leads to rapid increase in muscle strength • Used to <i>reverse</i> the neuromuscular block produced by non-depolarizing muscular blockers
Physostigmine	• Tertiary amine • <i>Can enter</i> and stimulate CNS	• Treatment of <i>overdoses</i> of anticholinergic drugs
Neostigmine	• Quaternary ammonium • <i>Does not</i> enter CNS	• Urinary retention • Reversal of effects of non-depolarizing neuromuscular blockers after surgery • Treatment of myasthenia gravis
Pyridostigmine	• Quaternary ammonium • Does not enter CNS	• Treatment of myasthenia gravis
Organophosphates	• Synthetic compounds • Many are extremely toxic	Insecticides: • Malathion • Sarin

ORGANOPHOSPHATES: INSECTICIDES

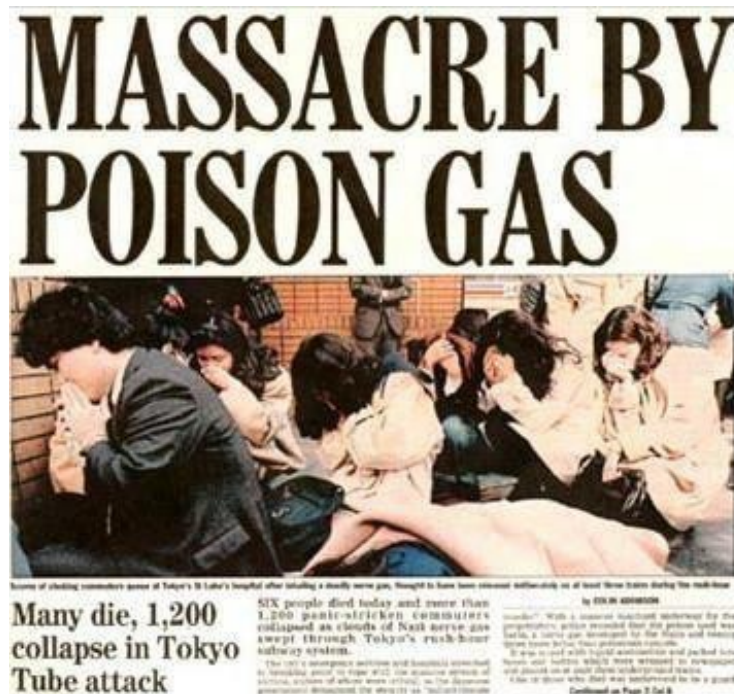
MALATHION



ORGANOPHOSPHATES: NERVE AGENTS

SARIN

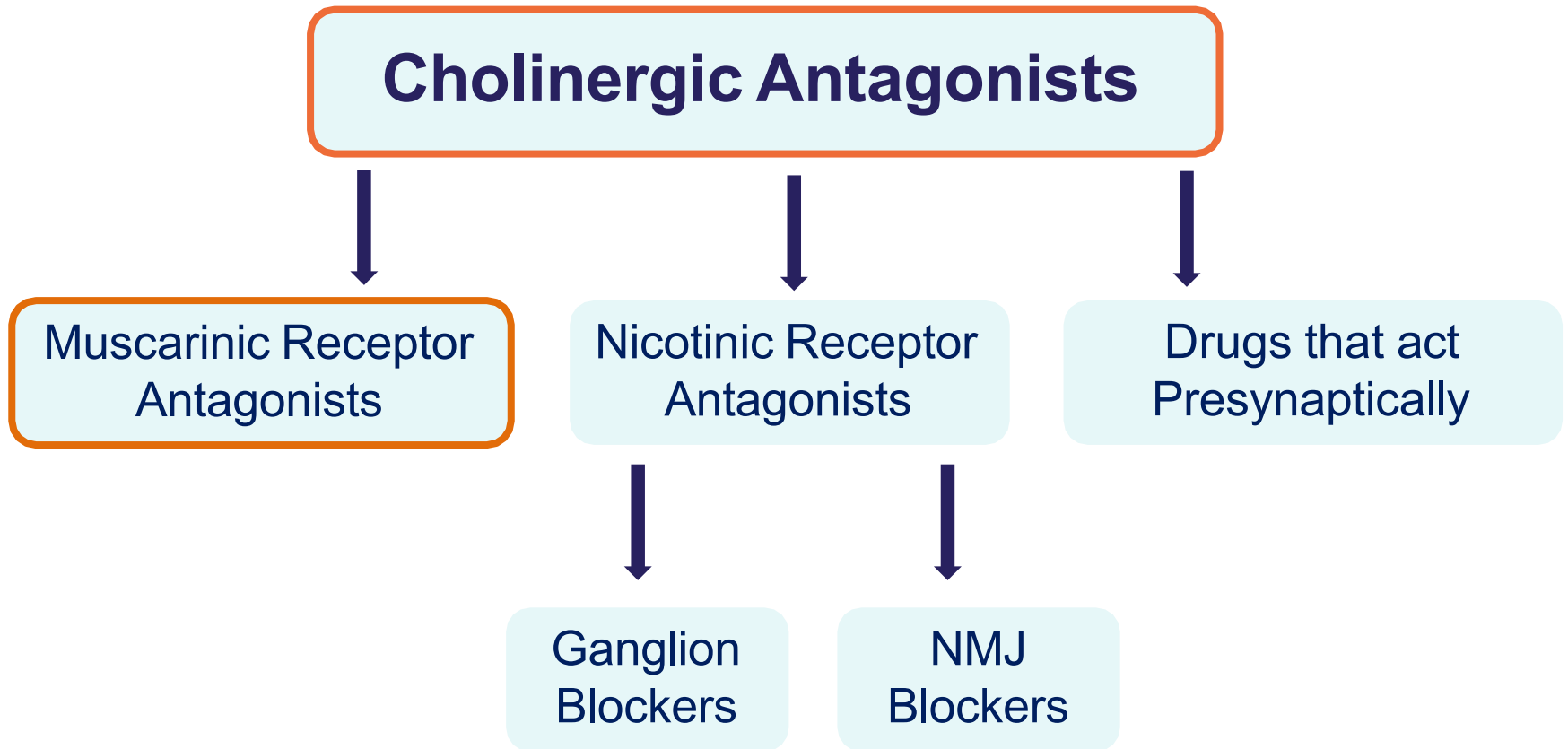
- Among the most potent synthetic toxic agents known.



CHOLINERGIC ANTAGONISTS



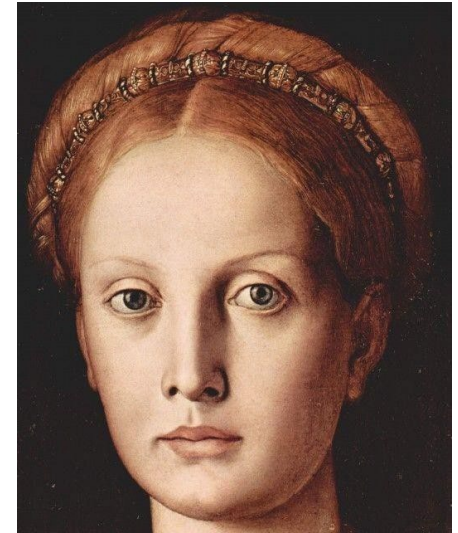
Cholinergic Antagonists



Muscarinic Receptor Antagonists: Belladonna Alkaloids (Atropine & Scopolamine)

Atropine

- **Prototype** of muscarinic antagonists
- **Reversible competitive antagonist** at muscarinic receptors
- Tertiary amine: both **central** and **peripheral** muscarinic blocker



Muscarinic Receptor Antagonists: Belladonna Alkaloids (Atropine & Scopolamine)

Atropine

Actions

- M3 blockade** • Eye: Mydriasis & cycloplegia
 - GI: Reduces gastric motility
 - Urinary system: Decreases hypermotility of urinary bladder
 - Secretions: Salivary, sweat and lachrymal glands are blocked
 - Inhibition of sweat glands may cause high body temperature
- Atrial M2 blockade** • CV system: Moderate to high therapeutic doses cause tachycardia

Belladonna Alkaloids

Atropine

- Uses:**
- Antidote for cholinergic agonists
 - Blocks respiratory tract secretions prior to surgery

Adverse Effects

- Dry mouth
- Blurred vision
- Sandy eyes
- Tachycardia
- Constipation
- Urinary retention
- Effects on CNS:
 - Restlessness
 - Confusion
 - Hallucinations
 - Delirium



Hot as a Hare
(↑ temperature, ↓ sweating)



Dry as a Bone
(decreased secretions,
thirsty)



Red as a Beet
(flushed face, tachycardia)



Mad as a Hatter
(confusion, delirium)

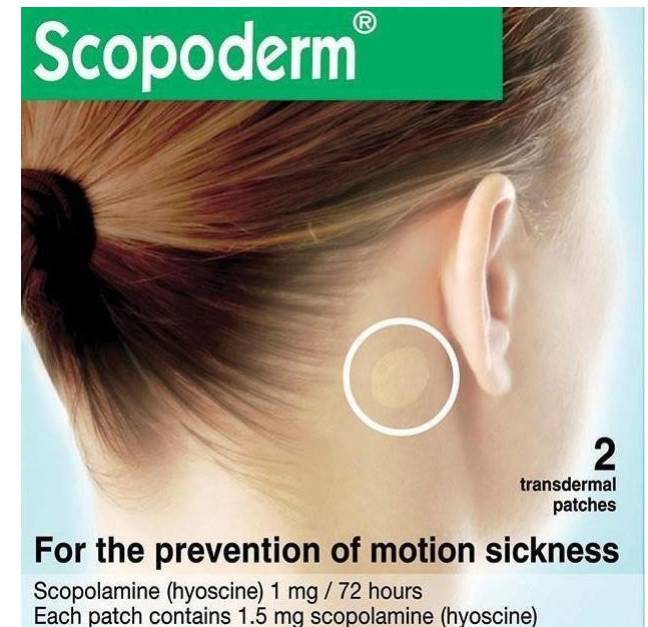
Belladonna Alkaloids

Scopolamine

Tertiary amine.

Produces peripheral effects similar to atropine but greater actions on CNS.

Uses: • Prevention of motion sickness



Synthetic & Semisynthetic Muscarinic Antagonists

Quaternary Ammonium Muscarinic Antagonists

Ipratropium

Action: Blocks M3 receptors in bronchial smooth muscle → causes bronchodilation

Uses: • Treatment of chronic obstructive pulmonary disease (COPD) & Asthma



Synthetic & Semisynthetic Muscarinic Antagonists

Tertiary Amine Muscarinic Antagonists

Tropicamide

Action: Blocks M3 receptors on iris sphincter and ciliary muscle

Uses:

- Mydriatic for fundoscopy
- Produces mydriasis with cycloplegia



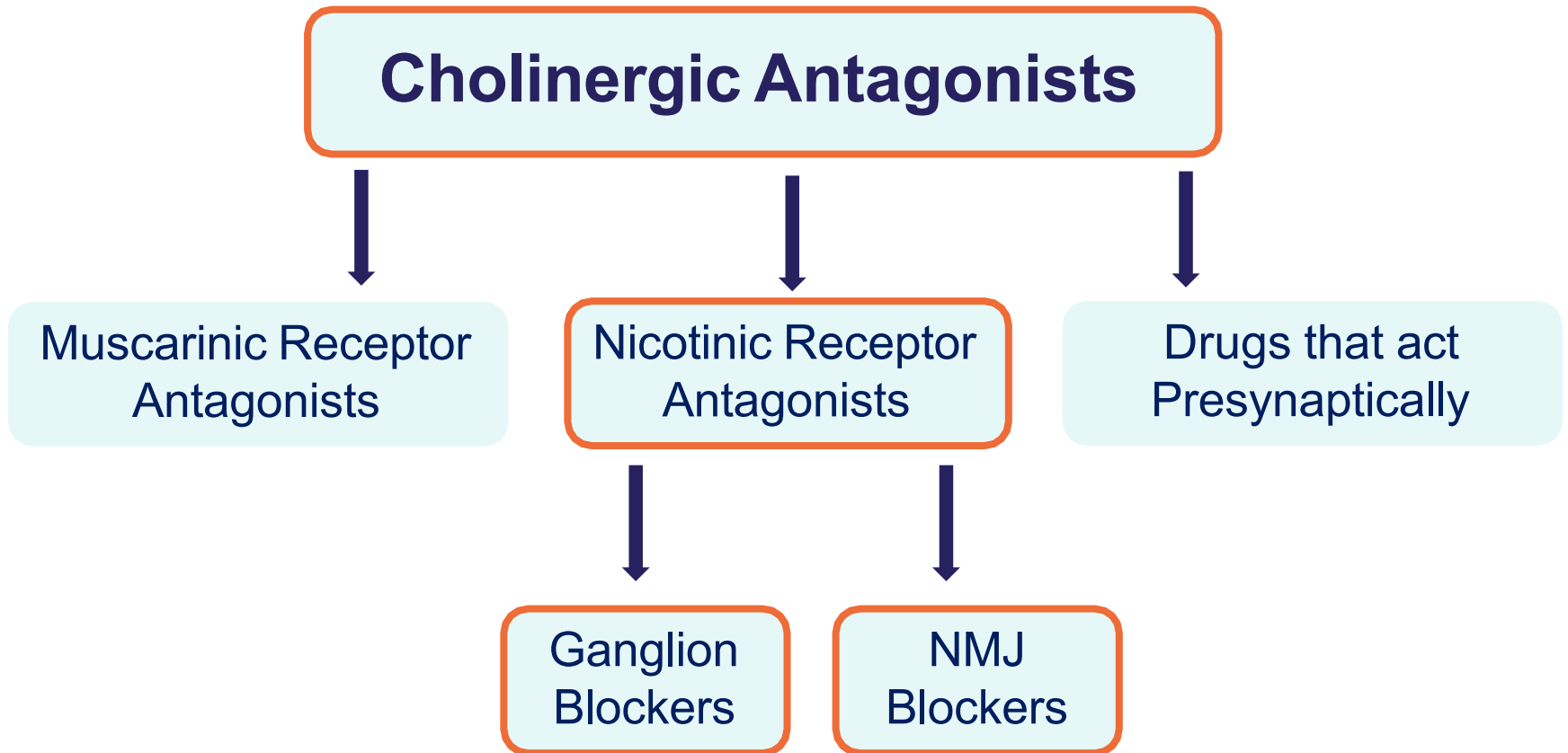
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Contraindications of Antimuscarinic Agents

- Contraindicated in patients with **angle-closure glaucoma**
- Should be used with caution in patients with **prostatic hypertrophy and in the elderly**



Cholinergic Antagonists



Nicotinic Receptor Antagonists

Ganglion Blockers

Ganglion blockade may occur by the following mechanisms:

- Prolonged depolarization (*Example*: Nicotine)
- Antagonism of nicotinic receptors (*Example*: Hexamethonium)

Uses: • Hexamethonium was used for hypertension in the past

- Due to their adverse effects, ganglion blockers have been *replaced* by **superior antihypertensive agents**

Nicotinic Receptor Antagonists

Ganglion Blockers

Pharmacological properties

- Effects of ganglion blockers can be predicted by a knowledge of which division of the ANS exercises **dominant control** of various organs
- The effect of the ganglion blocker is to **remove** the dominant control

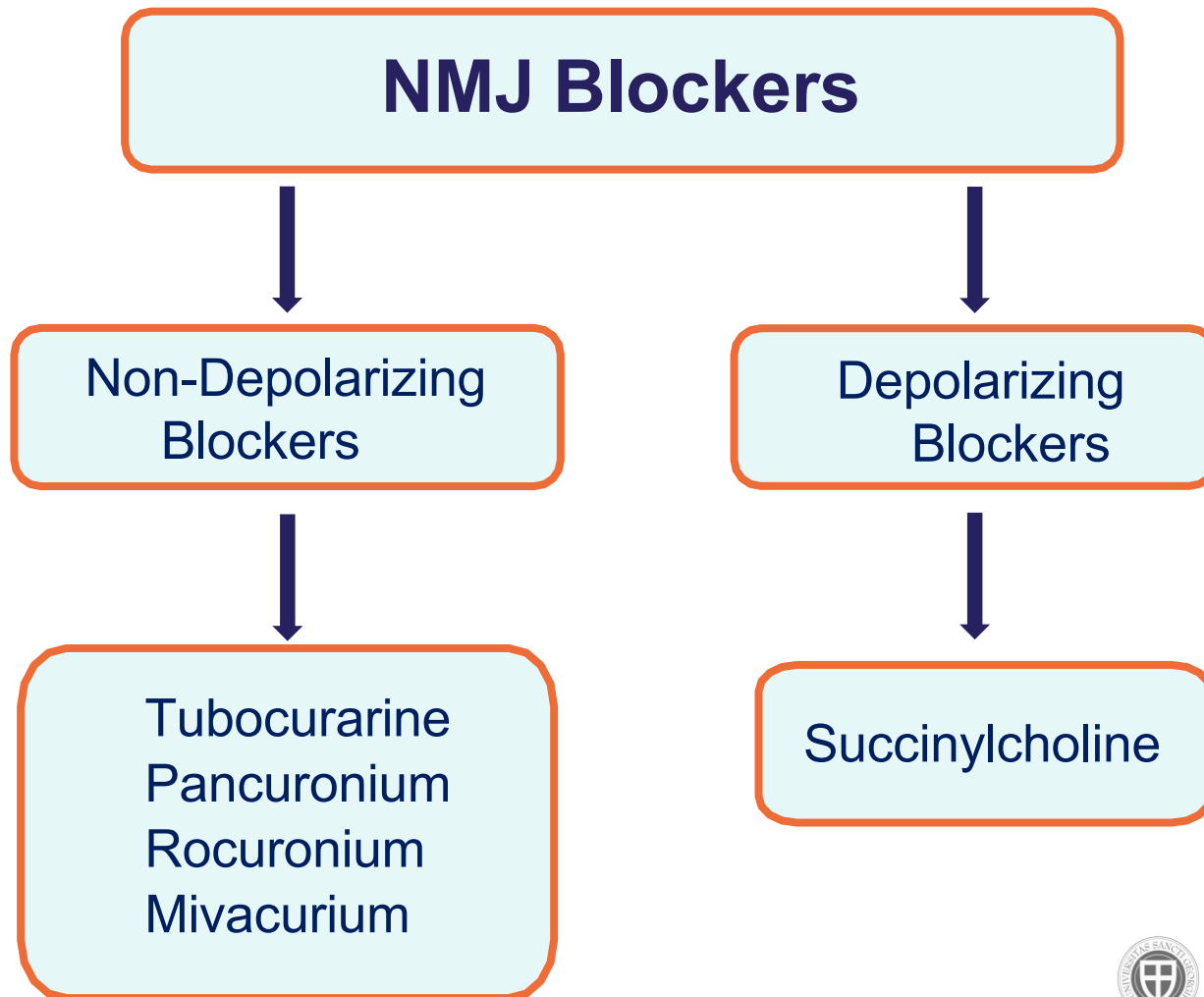
Nicotinic Receptor Antagonists

Effector Site	Predominant Tone	Receptor Type	Predominant Tone Effect
Arterioles	Sympathetic (<i>adrenergic</i>)	$\alpha 1$	Vasoconstriction; hypertension
Veins	Sympathetic (<i>adrenergic</i>)	$\alpha 1$	Venoconstriction
Heart	Parasympathetic (<i>cholinergic</i>)	M2	↓ Heart Rate
Iris	Parasympathetic (<i>cholinergic</i>)	M3	Miosis
Ciliary muscle	Parasympathetic (<i>cholinergic</i>)	M3	Accommodation to near vision
GI tract	Parasympathetic (<i>cholinergic</i>)	M3	↑ tone and motility; diarrhea; ↑ secretions
Urinary bladder	Parasympathetic (<i>cholinergic</i>)	M3	Urinary incontinence
Salivary glands	Parasympathetic (<i>cholinergic</i>)	M3	Hypersalivation
Sweat glands	Sympathetic (<i>cholinergic</i>)	M3	Hyperhidrosis

Nicotinic Receptor Antagonists

Effector Site	Predominant Tone	Receptor Type	Predominant Tone Effect	Effect of Ganglionic Blockade
Arterioles	Sympathetic (adrenergic)	$\alpha 1$	Vasoconstriction; hypertension	Vasodilation; Hypotension
Veins	Sympathetic (adrenergic)	$\alpha 1$	Venoconstriction	Venodilation
Heart	Parasympathetic (cholinergic)	M2	↓Heart Rate	↑Heart Rate
Iris	Parasympathetic (cholinergic)	M3	Miosis	Mydriasis
Ciliary muscle	Parasympathetic (cholinergic)	M3	Accommodation to near vision	Cycloplegia (focus for far vision)
GI tract	Parasympathetic (cholinergic)	M3	↑tone and motility; diarrhea; ↑secretions	↓ tone and motility; constipation; ↓ Secretions
Urinary bladder	Parasympathetic (cholinergic)	M3	Urinary incontinence	Urinary retention
Salivary glands	Parasympathetic (cholinergic)	M3	Hypersalivation	Xerostomia
Sweat glands	Sympathetic (cholinergic)	M3	Hyperhidrosis	Anhidrosis

Nicotinic Receptor Antagonists



NMJ Blockers – Non-depolarizing Blockers

Tubocurarine, Pancuronium , Rocuronium, Vecuronium Mivacurium

- Reversible competitive antagonists.
- **Tubocurarine is the prototype.**

Mechanism of Action

- Nondepolarizing blockers first cause motor weakness.
- Ultimately, skeletal muscles become totally flaccid and inexcitable to stimulation.
- Their action can be overcome by increasing the concentration of acetylcholine in the synapse with administration of **neostigmine or edrophonium**.

NMJ Blockers – Non-depolarizing Blockers

SHORT- ACTING	INTERMEDIATE- ACTING	LONG-ACTING
Mivacurium	Rocuronium Vecuronium	Tubocurarine Pancuronium

- Hydrolysis by plasma butyrylcholinesterase is the primary mechanism for inactivation.
- Metabolism not dependent on liver or kidney.

NMJ Blockers – Non-depolarizing Blockers

Adverse Effects

- In general, these agents are safe with minimal side effects.
- Tubocurarine may release histamine and block ganglionic Nn receptors.
- Replaced by new drugs with fewer adverse effects.



NMJ Blockers –Depolarizing Blockers

Succinylcholine

- Succinylcholine activates the Nm receptor and depolarizes the Junction → causes fasciculations.
- Succinylcholine is not metabolized effectively by acetylcholinesterase → leads to channel inactivation and flaccid paralysis.
- The onset of neuromuscular blockade is very rapid, usually <1 minute.
- The duration of the block is approximately 5 minutes due to hydrolysis by plasma cholinesterases.

NMJ Blockers –Depolarizing Blockers

Succinylcholine

Adverse Effects

- Muscle Pain
- **Hyperkalemia**
 - Due to **loss of tissue potassium** during depolarization.
 - Risk of hyperkalemia is higher in patients with burns or muscle trauma.
 - Hyperkalemia may lead to arrhythmias, cardiac arrest or circulatory collapse.
- Increased Intraocular Pressure

NMJ Blockers –Depolarizing Blockers

Succinylcholine

Adverse Effects

- **Malignant Hyperthermia**

- Succinylcholine can induce malignant hyperthermia in genetically susceptible patients → Most cases are due to a defect in the ryanodine receptor gene (RYR1).
- Malignant hyperthermia is one of the main causes of death due to anesthesia.
- Due to excessive release of Ca^{2+} from the SR.
- The syndrome includes tachycardia and hypertension, severe muscle rigidity, hyperthermia, hyperkalemia, and acidosis.
- **Treatment: dantrolene. Blocks release of Ca^{2+} from SR.**

NMJ Blockers –Depolarizing Blockers

Succinylcholine

Contraindications

- History of malignant hyperthermia
- History of skeletal muscle myopathies.
- Major burns.
- Multiple trauma
- Denervation of skeletal muscle
- Upper motor neuron injury.

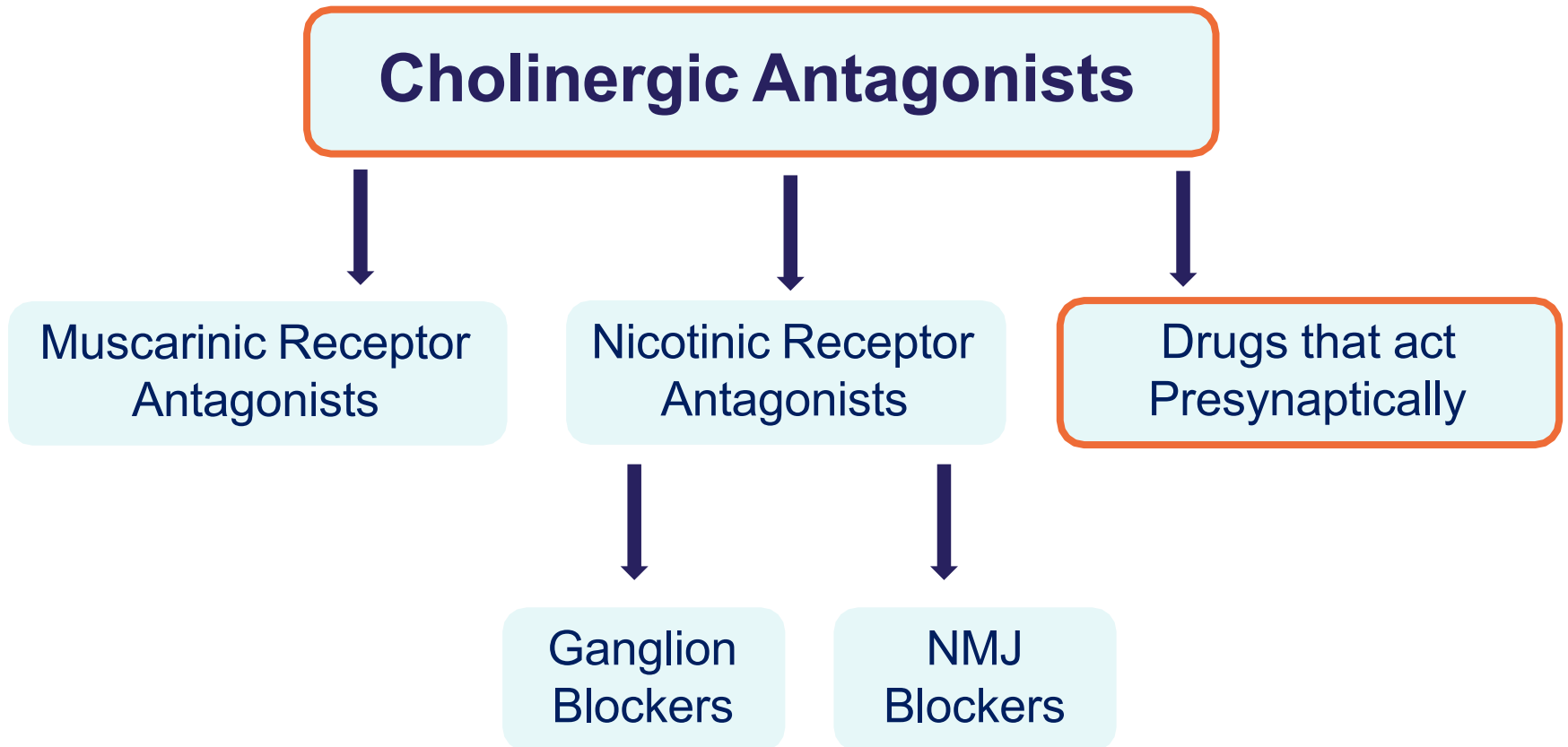
NMJ Blockers

Uses

- **Skeletal Muscle Relaxation During Anesthesia**
 - Main clinical use: adjuvants in surgical anesthesia to obtain relaxation of skeletal muscle.
 - **Succinylcholine** is used to facilitate endotracheal intubation during induction of anesthesia. Also used during ECT.
- **Reversal of Nondepolarizing Neuromuscular Blockade**
 - Upon completion of a surgical procedure neostigmine or edrophonium can be given to reverse competitive blockade.
 - Atropine is given concomitantly to prevent bradycardia.



Cholinergic Antagonists



Drugs That Act Presynaptically

Inhibitors of Acetylcholine Release

Botulinum Toxin

- Uses**
- Injected locally into muscles for treatment of severe diseases involving muscle spasms
 - Approved for cosmetic treatment of facial wrinkles



Cholinergic Drugs List

Cholinergic Agonists

Direct acting drugs

- Acetylcholine
- Bethanechol
- Methacholine
- Pilocarpine
- Nicotine

AchE inhibitors

- Edrophonium
- Physostigmine
- Neostigmine
- Pyridostigmine
- Malathion
- Sarin

Cholinergic Antagonists

Muscarinic antagonists

- Atropine
- Scopolamine
- Ipratropium
- Tropicamide

Ganglion blockers

- Hexamethonium

Neuromuscular blockers

- Tubocurarine
- Pancuronium
- Rocuronium
- Mivacurium
- Succinylcholine

Drugs that act presynaptically

- Botulinum Toxin



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